Does a Bigger Discount Mean a Better-Rated Product?

MATH563 Project - Amazon India Products Analysis

Megan Millet

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Research Question & Dataset

Dataset: Amazon India product listings, Kaggle 2023
1,426 products across three categories after cleaning

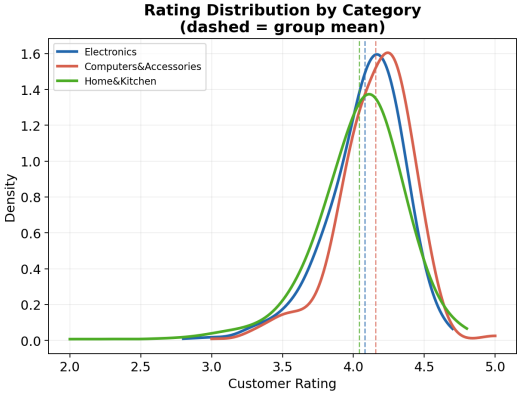
Category	n	Share
Electronics	526	36.9%
Computers&Accessories	453	31.8%
Home&Kitchen	447	31.4%

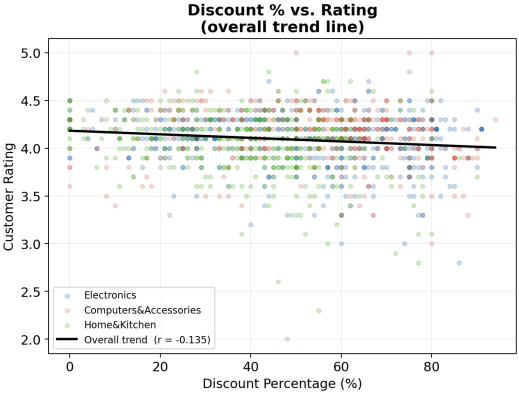
Research question:

Does a product's discount percentage predict its customer rating, and do mean ratings differ across product categories?

Methods: Confidence intervals · One-way ANOVA · OLS regression (Model M2)

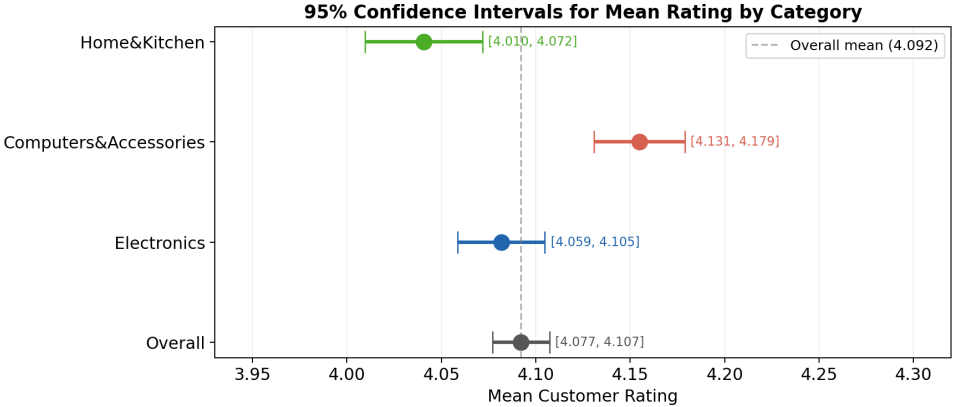
Exploratory Overview





Ratings cluster tightly near 4.0–4.3 · Overall trend: $r = -0.135$, $p < 0.001 \implies$ bigger discounts, *slightly lower* ratings

95% Confidence Intervals for Mean Rating



Computers&Acc [4.131, 4.179] and Home&Kitchen [4.010, 4.072] **do not overlap** — strong visual evidence of a real difference

One-Way ANOVA: Rating by Category

ANOVA Results

Source	SS	df	MS	F
Between (Category)	0.629	2	0.314	18.15
Within (Residual)	24.657	1423	0.0173	
Total	25.286	1425		

$p < 0.001 \implies$ Reject H_0: category means are not all equal.

Post-hoc Pairwise Comparisons (Bonferroni-adjusted, $\alpha = 0.0167$)

Pair	Mean Difference	Verdict
Computers&Acc vs Home&Kitchen	$+0.114$ stars	**Sig.**
Electronics vs Computers&Acc	-0.073 stars	**Sig.**
Electronics vs Home&Kitchen	$+0.041$ stars	n.s.

Why We Control for Category

The confound: Home&Kitchen has both lower discounts *and* lower ratings

Category	Mean discount	Mean rating
Computers&Accessories	54.0%	4.155
Electronics	50.8%	4.082
Home&Kitchen	40.2%	4.041

Without controlling for category, the discount effect is **understated**.

Model M2 separates the two:

$$\widehat{\text{Rating}} = \beta_0 + \beta_1 \cdot \text{Discount\%} + \beta_2 \cdot \mathbf{1}[\text{Electronics}] + \beta_3 \cdot \mathbf{1}[\text{Home\&Kitchen}]$$

The discount slope strengthens from -0.0019 *(uncontrolled) to* -0.0026 *(controlled).*

Model M2 — What the Coefficients Mean

Estimated model

$$\widehat{\text{Rating}} = 4.295 - 0.0026 \cdot \text{Discount\%} - 0.082 \cdot \mathbf{1}[\text{Electronics}] - 0.150 \cdot \mathbf{1}[\text{Home\&Kitchen}]$$

Interpretation (holding other terms fixed)

- **Discount slope (-0.0026):** each $+1$ percentage point discount predicts a 0.0026-star lower rating
- **Electronics (-0.082):** Electronics products rate 0.082 stars below Computers&Accessories at the same discount
- **Home&Kitchen (-0.150):** Home&Kitchen products rate 0.150 stars below Computers&Accessories at the same discount

Reference category is Computers&Accessories.

Model M2 — Significance and Practical Size

Statistical result

- All three predictors are statistically significant ($p < 0.001$)
- The negative discount effect remains after controlling for category

Practical size check

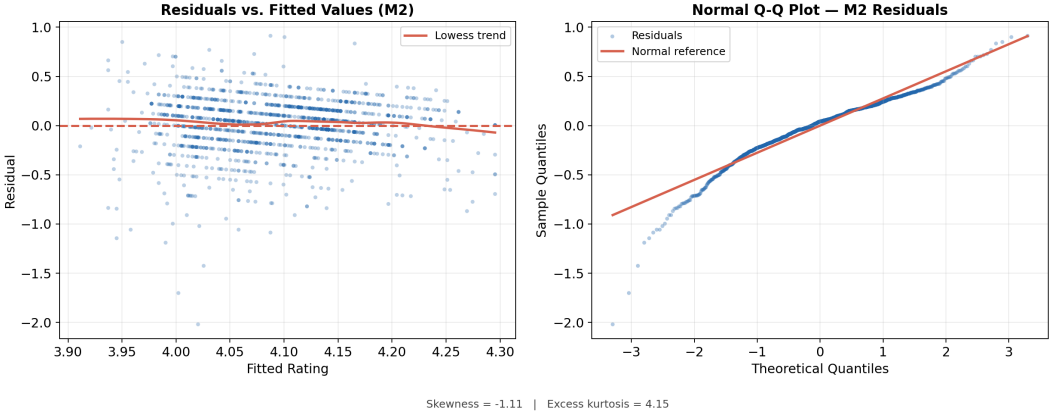
$$\Delta\widehat{\text{Rating}} = (90 - 10) \times (-0.0026) = 80 \times (-0.0026) = -0.208$$

A product at 90% discount versus 10% discount in the same category is predicted to rate about 0.21 stars lower.

Model fit

- $R^2 = 0.057$: discount and category explain about 5.7% of rating variation
- **Why?** My Take: Most variation is still due to unobserved factors (quality, brand, review composition)

Model Diagnostics (M2)



Residuals vs. Fitted: no systematic pattern, variance stable · Q-Q: moderate left-skew expected from bounded $[1, 5]$ ratings; looks decently reliable

Why Are Bigger Discounts Associated with Lower Ratings?

The negative relationship ($r = -0.135$, M2 slope $= -0.0026$) holds across all categories.

Three plausible explanations:

1. **Quality clearing**: vendors discount slow-moving, lower-quality inventory; buyers notice
2. **Expectation inflation**: shoppers who feel they got a "deal" hold higher expectations, then rate more harshly
3. **Brand confounding**: budget brands discount more aggressively *and* rate lower; discount acts as a proxy for quality tier

Note: This is observational data. We cannot conclude that reducing discounts would raise ratings in general.

Conclusions & Limitations

Three take-aways:

1. **Category matters more than discount.** Computers&Acc rates 0.11 stars above Home&Kitchen, so it is significant and practically noticeable.
2. **Bigger discounts** $\Rightarrow$ **slightly lower ratings**; opposite of the intuitive expectation. Effect is real but small: 80 pp extra discount ≈ -0.21 stars.
3. **Weak overall fit** ($R^2 \approx 6\%$): product quality, branding, and review composition dominate but are hidden/unobserved variables.

Limitations:

- No causal identification
- Hidden/omitted variables: branding, product age, quality, etc., are unobserved but have large effects on rating
- Ratings are bounded $[1, 5]$ and left-skewed; OLS assumptions only approximately met
- Amazon India 2023 only; these findings do not generalize to other markets or time periods